Comparison of EKF and UKF State Estimation with Measurement Gating and RK4 Propagation

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***Abstract*—This report addresses the six required tasks in the EMCH792 final project for the nonlinear kinematic bicycle model. The calibration pass yielded R_1 = 0.087247 and R_2 = 0.006536 with a constant sample period of 0.100 s. The vanilla rectangular comparison favored the UKF over the EKF, reducing position RMSE from 3.6891 m to 1.6879 m and heading RMSE from 17.8381 deg to 4.9532 deg. A fixed scalar 99% normalized-innovation-squared gate with threshold 6.6349 improved heading RMSE by 12.7993 deg for the EKF and 1.1132 deg for the UKF, but position RMSE increased because repeated y_1 rejections made the gated runs more prediction dominated. Replacing rectangular UKF propagation with RK4 reduced position RMSE by 0.2673 m, while the RK4 runtime increased by only 0.950 ms relative to the rectangular UKF.**

***Index Terms*—Extended Kalman filter, Unscented Kalman filter, Nonlinear state estimation, Measurement gating, Runge-Kutta integration**

I. Problem I. Calibration Variance Estimation

The first task was to estimate the two sensor variances from the calibration data and confirm the deployment sample period. The calibration pass produced R_1 = 0.087247 for the range-like measurement and R_2 = 0.006536 for the cubic heading sensor. The deployment dataset remained uniformly sampled at 0.100 s. The filters were initialized from the first truth sample and used an initial covariance equal to 0.50 of the nominal base matrix so that the baseline comparison started from the same prior in every case.

II. Problem II. Vanilla EKF and UKF with Rectangular Integration

Both filters used the assignment state $x = [x, y, \psi, u]^T$ with steering angle and longitudinal acceleration as inputs, fixed geometry $l_f = l_r = 1$, and the prescribed continuous-time process covariance. The EKF used rectangular Euler prediction with first-order Jacobian linearization for both the process and measurement updates. The UKF used the scaled unscented transform with $\alpha = 10^{-}3$, $\beta = 2$, and $\kappa = 0$, and propagated every sigma point with the selected discrete-time integrator. In every configuration, the two measurements were processed sequentially as scalar updates so that the gating study in Problem IV could test each sensor independently.

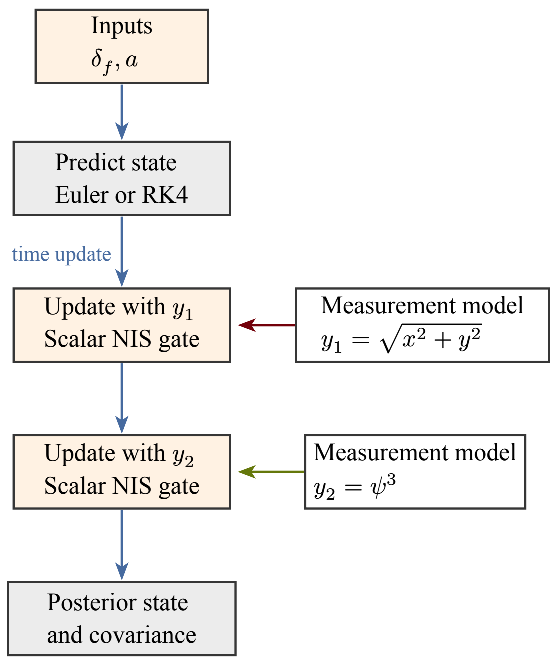

Fig. 1. Standalone CeTZ diagram of the sequential estimation workflow. The figure was authored separately, compiled to PDF, and then imported into the main IEEE-formatted document as a standard image asset.

The ungated rectangular comparison favored the UKF. The EKF rectangular run produced a position RMSE of 3.6891 m with a heading RMSE of 17.8381 deg, while the rectangular UKF reduced these values to 1.6879 m and 4.9532 deg. The same trend appeared in the covariance histories. The rectangular EKF carried an average covariance trace of 0.7417, while the rectangular UKF carried 6.0709. The UKF therefore remained less overconfident on this nonlinear problem, especially under the cubic heading measurement.

III. Problem III. Runtime of the Vanilla EKF and UKF

The average runtime study for the vanilla rectangular filters used sixty repeated executions after one warm-up pass per configuration. The EKF remained the cheaper estimator at 2.228 ms per run on average, while the rectangular UKF required 3.976 ms. That extra cost is expected because the

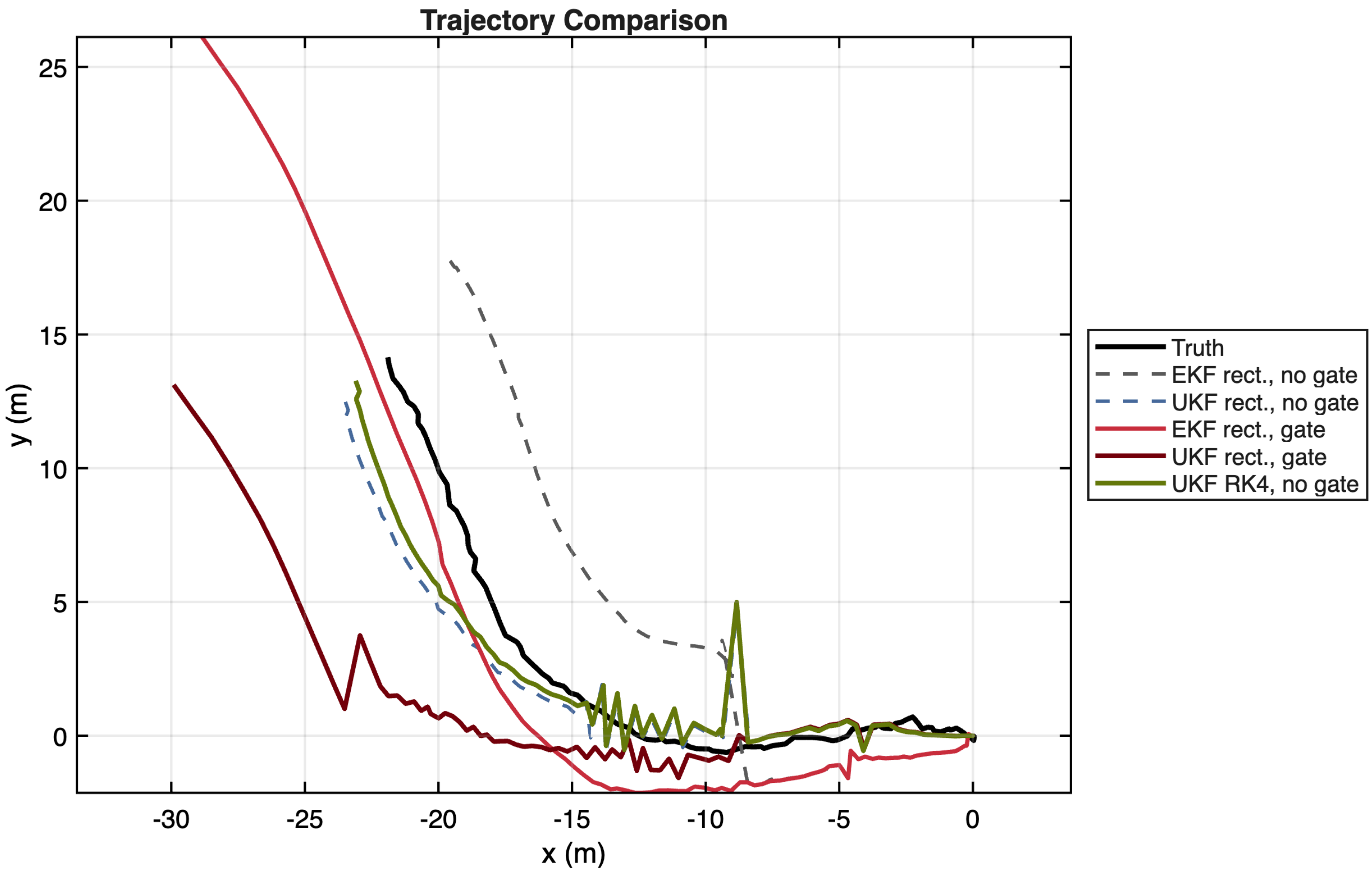

Fig. 2. Trajectory comparison for the rectangular EKF and UKF baselines, the gated rectangular filters, and the ungated RK4 UKF. The vanilla UKF and the ungated RK4 UKF remain closest to the truth trajectory, while the gated runs drift late in the record after repeated measurement rejection.

UKF must propagate and recombine the sigma-point set at every step.

TABLE I
AVERAGE RUNTIME OVER SIXTY REPEATED EXECUTIONS FOR THE VANILLA RECTANGULAR EKF AND UKF.

Configuration	Mean ms	Std ms
EKF rect., no gate	2.228	0.846
UKF rect., no gate	3.976	0.329

IV. Problem IV. Measurement Gating

Measurement gating was implemented as a scalar normalized innovation squared test on each sensor separately, exactly as requested in the handout. The study used a fixed 99% gate with threshold 6.6349. The gate did what it was supposed to do for the heading-like sensor. The EKF heading RMSE improved by 12.7993 deg and the UKF heading RMSE improved by 1.1132 deg after the strongest y_2 outliers were rejected. However, the position RMSE increased by 0.2061 m for the EKF and by 2.4006 m for the UKF because the gate also rejected many late y_1 measurements. The final rectangular gated runs rejected 24 and 14 range measurements, respectively, which made both filters coast too long on the process model alone.

V. Problem V. UKF with Runge-Kutta Integration

Replacing the rectangular UKF propagation with RK4 improved the ungated UKF accuracy. The rectangular UKF produced a position RMSE of 1.6879 m, while the ungated RK4 UKF reduced that value to 1.4206 m for an improvement of 0.2673 m. The heading RMSE remained nearly unchanged at 4.9532 deg for the rectangular UKF and 5.0272 deg for the RK4 UKF. The main benefit therefore appeared in the planar trajectory and state histories, where the higher-order integrator reduced accumulated discretization error.

VI. Problem VI. Runtime of the RK4 UKF Versus the Rectangular UKF

The integrator runtime comparison again used sixty repeated executions after one warm-up pass per configuration. The rectangular UKF required 3.786 ms per run on average, while the RK4 UKF required 4.736 ms. The RK4 propagation therefore added 0.950 ms per run relative to the rectangular UKF. That extra cost is small compared with the position-accuracy gain reported in Problem V.

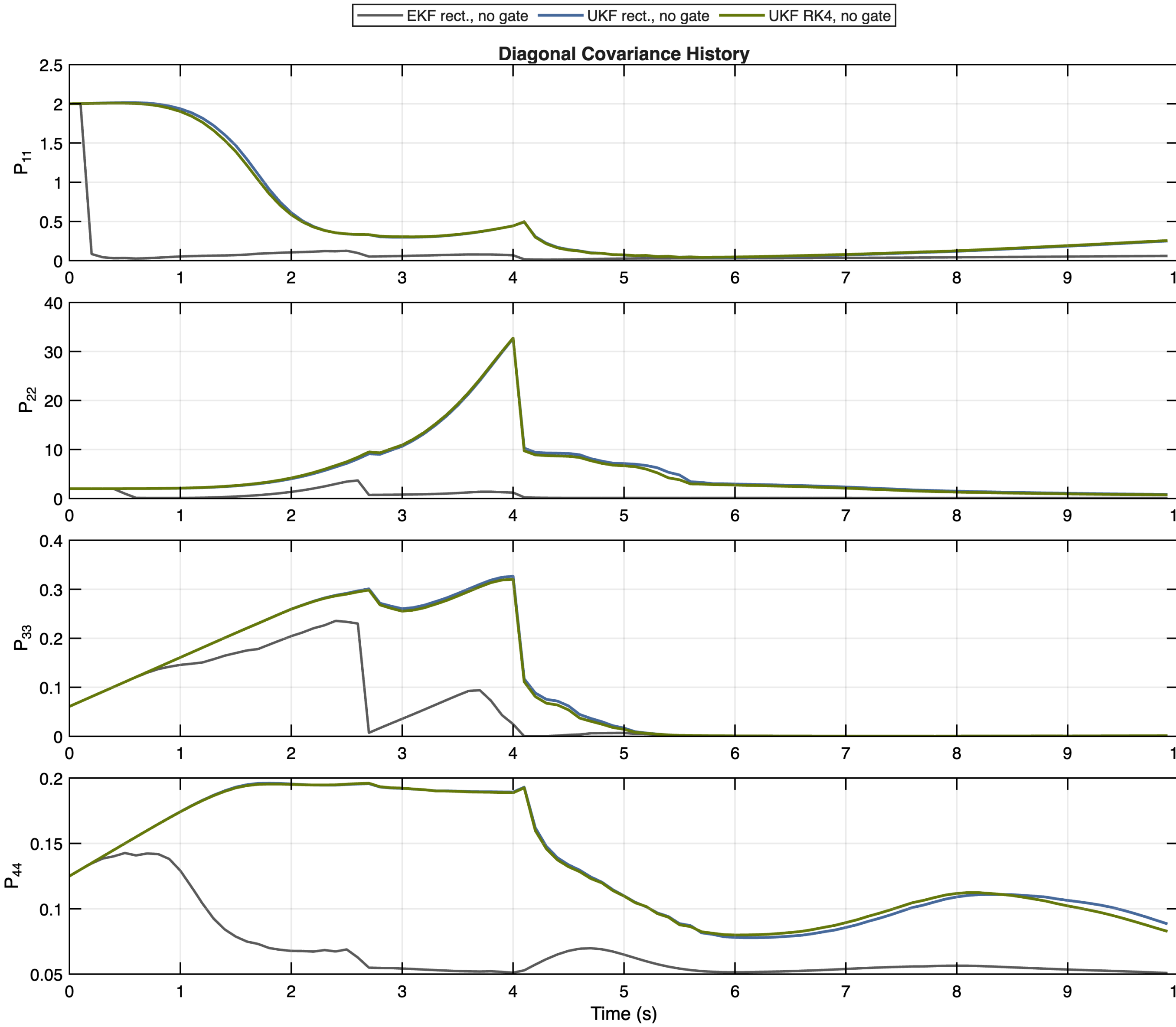

Fig. 3. Diagonal covariance histories for the rectangular EKF, rectangular UKF, and ungated RK4 UKF. The UKF variants retain visibly larger uncertainty than the EKF, which is consistent with their stronger nonlinear treatment.

TABLE II
AVERAGE RUNTIME OVER SIXTY REPEATED EXECUTIONS FOR THE RECTANGULAR AND RK4 UKF IMPLEMENTATIONS.

Configuration	Mean ms	Std ms
UKF rect., no gate	3.786	0.058
UKF RK4, no gate	4.736	0.017

Taken together, the six required comparisons point to the same conclusion. The vanilla UKF was the stronger baseline on this nonlinear problem, the scalar gate was useful for removing the worst heading-sensor outliers but harmed position accuracy when range updates were rejected too often, and RK4 was the cleanest algorithmic upgrade because it improved the ungated UKF trajectory with only a modest runtime increase.

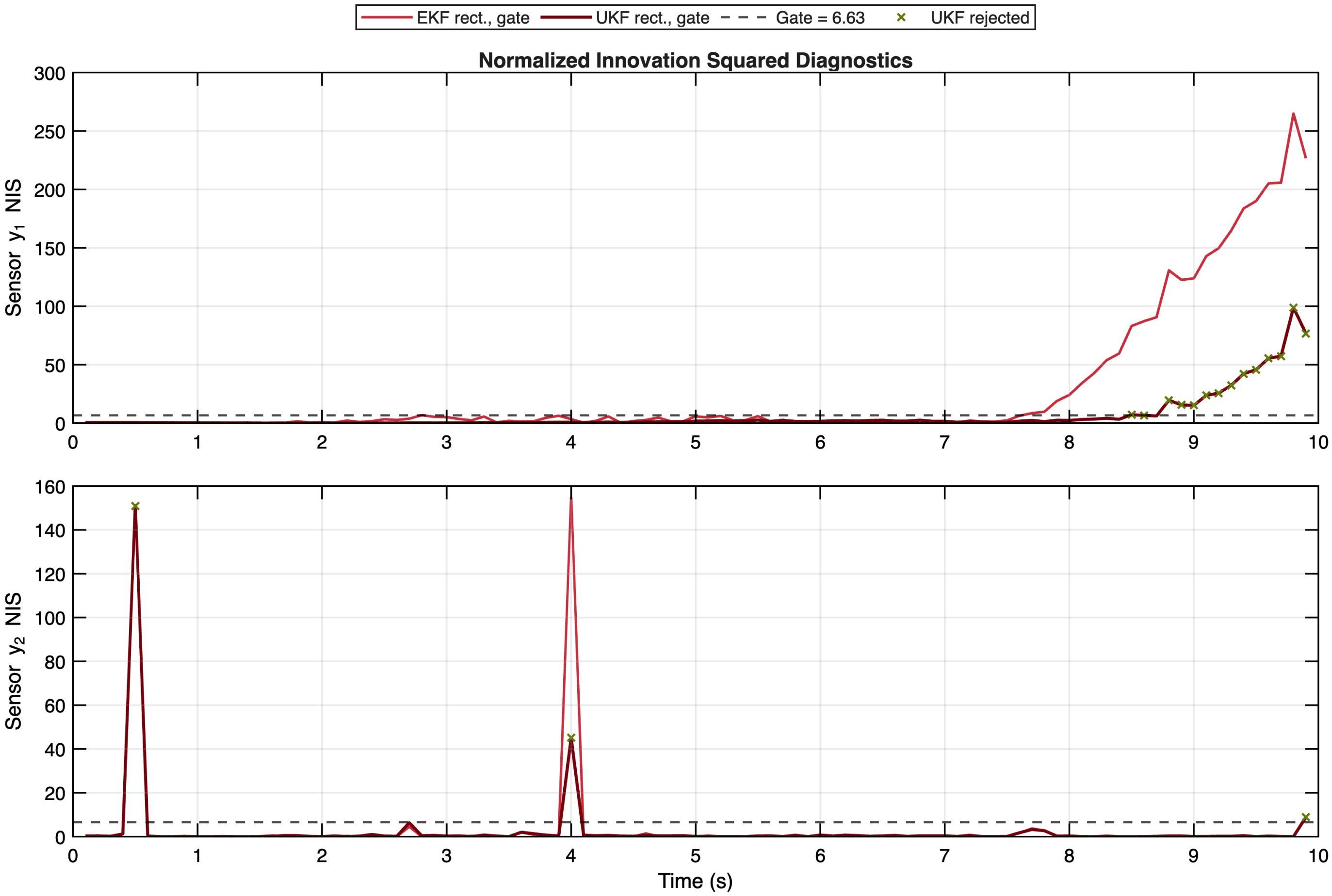

Fig. 4. Normalized innovation squared histories for both scalar sensors. The isolated y_2 spikes are clearly detectable, while the late growth in y_1 residuals explains the large number of range-measurement rejections.

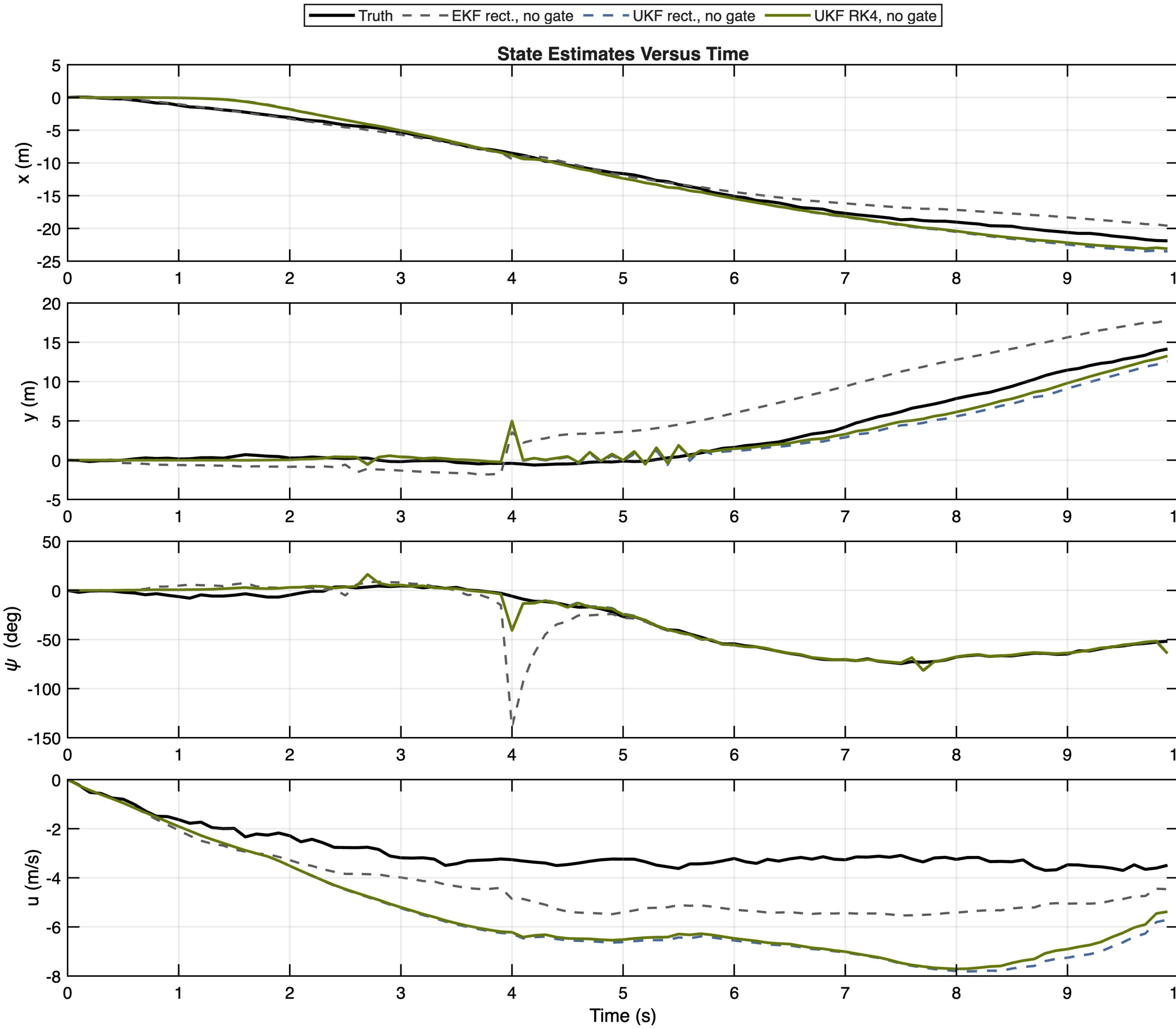

Fig. 5. State estimates versus time for the vanilla rectangular EKF, vanilla rectangular UKF, and ungated RK4 UKF. The RK4 UKF mainly improves the position channels while leaving the heading history close to the rectangular UKF result.

TABLE III
State-estimation accuracy, covariance summary, and measurement-rejection counts for all evaluated configurations.

Configuration	x	y	psi deg	u	pos.	avg tr(P)	rej y1	rej y2
EKF rect., no gate	1.1828	3.4943	17.8381	1.5688	3.6891	0.7417	0	0
UKF rect., no gate	1.0710	1.3046	4.9532	2.9613	1.6879	6.0709	0	0
EKF rect., gate	1.8305	3.4382	5.0388	3.6671	3.8951	1.1537	24	3
UKF rect., gate	2.2324	3.4252	3.8400	4.5974	4.0885	13.0001	14	3
UKF RK4, no gate	0.9634	1.0441	5.0272	2.8578	1.4206	5.9494	0	0
UKF RK4, gate	2.2670	2.6936	3.7895	4.5681	3.5206	12.7821	15	3